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4.2.4. Most Frequently Sold Product Pairs

01:25

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

**Sample Data :**

Date,Product,Quantity,Price,City

2025-01-01,Product **A**,5,20,New York

2025-01-01,Product **B**,3,15,Los Angeles

2025-01-02,Product **A**,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product **B**,2,15,Chicago

2025-01-03,Product **A**,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product **B**,5,15,Los Angeles

2025-01-05,Product **A**,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

**Explanation:**

**Transactions:**

- **2025-01-01:** Product A, Product B

- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A
- **2025-01-04:** Product C, Product B
- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B** : Appear in transactions on 2025-01-01 and 2025-01-03.
- **Product A and Product C** : Appear in transactions on 2025-01-02 and 2025-01-05.
- **Product B and Product C** : Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

**Note:**

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)
# write the code
pair_counter = Counter()

# Group by Date and generate product pairs
for _, group in df.groupby('Date'):
    products = group['Product'].tolist()
    pairs = combinations(sorted(products), 2)
    pair_counter.update(pairs)

# Find maximum frequency
max_count = max(pair_counter.values())

# Output the most frequent product pairs
for pair, count in pair_counter.items():
    if count == max_count:
```

```
print(f"{pair[0]} and {pair[1]}: {count} times")
```